CHANDERNAGORE COLLEGE

SEMESTER – IV B.A. / B.Sc. / B.COM (MAJO R/MINOR/SEC) INTERNAL ASSESSMENT EXAMINATION, 2025

SUBJECT: Computer Science COURSE CODE: COMP4013

COURSE NAME: Database Management System

FULL MARKS: 10 TIME: 30 MINUTES

ATTEMPT ANY 01 OUT OF THE FOLLOWING 03 QUESTIONS

1.a. Explain the three-schema architecture of a DBMS. Describe each level (external, conceptual, and internal schema and explain how they relate to the concepts of logical and physical data independence.

b. Explain the SQL instruction FULL OUTER JOIN with the help of Relational Algebra notations.

2.a. Discuss the concept of normalization in database design. Explain the purpose of normalization and describe the differences between the first, second, and third normal forms (1NF, 2NF, 3NF). Provide a simple example of a table that violates 2NF and show how to normalize it.

b. Translate the following SQL query into its equivalent Relational Algebra expression. Explain each part of your relational algebra expression and how it corresponds to the original SQL query.

```
i.   SELECT COUNT(E.EName), D.DeptName
ii.  FROM Employee AS E JOIN Department AS D ON E.DeptID = D.DeptID
iii. WHERE E.Salary > 50000
iv.  GROUP BY D.DeptName
v.   HAVING COUNT(E.EName) >=5;
```

Describe the sequence of execution of statements marked by (i, ii iii, ...).

3.a. Explain the ACID properties of a database transaction. Define each of the four properties—Atomicity, Consistency, Isolation, and Durability—and provide a simple, real-world example to illustrate how all four properties ensure the reliability and integrity of a transaction.

b. Explain the concept of an attribute closure. Given a relation R(A, B, C, D, E) with the set of functional dependencies F = {A → BC, CD → E, B → D, E → A}, calculate the attribute closure of {B}, denoted as $\{B\}^+$. Show your steps clearly. Find the candidate keys for this relation R.